Bhartiya Vidya Bhavan's

Sardar Patel Institute of Technology

Department of Computer Engineering

Academic Year 2026-27

Mini Project - Weekly Progress Report

Mini Project Title: Dynamic Timetable: A Role-Based Timetable Monitoring System with Automated Timetable Extraction and Natural-Language Queries

Name of Students:
1. Shivam Mulani (UID 2024300141)
2. Gurshaan Nandrajog (UID 2024300146)
3. Harshvarddhan Waghela (UID 2024300272)

Name of the Supervisor: Prof. Anant Nimkar

Review Date	Details of Work Done and Review of Meeting	Students Signature	Remarks by Supervisor	Supervisor Signature
	Review 1: Problem identification and topic finalisation • Studied the department's published 10-page timetable PDF and listed the problems students and faculty face (manual lookup, lab batches, legends, late clash discovery, multiple versions). • Finalised the title and scope: read the published PDF, role-based views, plain-English questions, clash detection. • Work split agreed: Shivam (architecture, database, security, query engine), Gurshaan (extraction, clash detection, performance), Harshvarddhan (mobile/web app, testing, documentation).			

Review Date	Details of Work Done and Review of Meeting	Students Signature	Remarks by Supervisor	Supervisor Signature
	Review 2: Literature survey and requirements • Surveyed papers on university timetabling, table recognition/OCR and natural-language interfaces to databases; compared FET/UniTime/aSc, Tabula/Camelot and LLM text-to-SQL and identified the gap. • Prepared the SRS: functional requirements for 5 roles (Student, Professor, HOD, Principal, Admin) and non-functional targets (accuracy >= 95%, answer < 300 ms).			
	Review 3: System design • Designed the architecture (Expo app, FastAPI back end, background worker, PostgreSQL), DFDs, use-case and sequence diagrams. • Designed the database schema (31 tables, 3NF) and the REST API; finalised the technology stack.			
	Review 4: Back-end foundation • Shivam: schema and Alembic migrations, Argon2id login, JWT with rotating refresh tokens, role-based access control, admin master data. • Gurshaan: file intake for PDF, scanned image, Excel and Word. Harshvarddhan: app project set-up and screen flow.			
	Review 5: Extraction engine and clash detection • Gurshaan: grid extraction with PyMuPDF and Tesseract OCR, lab-cell splitting, page-wise legends, background worker; tuned on the real college PDF (noon-row label, combined B.E. page). • Time-row backtracking algorithm and group-sort-sweep clash detection: 542 raw overlaps reduced to the 2 genuine clashes.			
	Review 6: Query engine and mobile app • Shivam: natural-language query engine with entity resolution and clarification (e.g. 'at 2' asks AM/PM), parameterised SQL templates. • Harshvarddhan: Expo app with role dashboards, weekly timetable, Ask screen, upload, review and admin screens; Jest and end-to-end tests.			

Review Date	Details of Work Done and Review of Meeting	Students Signature	Remarks by Supervisor	Supervisor Signature
	Review 7: Real-data testing and deployment • Seeded the real timetable with one student login per lab group; 452 of 453 classes verified (99.8%). • App runs on Android, iPhone (Expo SDK 58) and laptop browser; free online demo on Render; fixed a password-reset security issue found in the audit. • Measured performance: 37-47 ms median answer time, 0 wrong answers with 32 parallel clients.			
	Review 8: Report and poster • Wrote the mini project report in LaTeX (SPIT format), designed the A3 poster and prepared the ESE demo guide; recorded each member's contribution.			
	Review 9: Improvements suggested in review • Replaced the fixed list of ~30 question types with a compositional query planner (17 operations, filters, dates, time windows, sorting). • Teachers can be found by code, first name, surname or 'X sir'; subject typos matched; everyone can view any teacher's, batch's or room's timetable while personal timetables stay private.			
	Review 10: Final submission and ESE preparation • Research paper in IEEE format, 29-slide presentation, scenario-based challenges added to the report; report re-formatted to the department's Overleaf template. • 179 back-end + 20 app tests and 45 end-to-end checks passing; plagiarism and AI-content reports to be generated; demo rehearsal.			